

# MARÍA MILAGROS LAHUERTA

Biomedical Engineering Master's Student

@ milagroslahuerta00@gmail.com +39 331 806 7506

Turin, Italy 15/08/2000 MARÍA MILAGROS LAHUERTA



## PROFILE

Final-year Master of Science student in Biomedical Engineering, completing a double degree program between the National University of Córdoba and Politecnico di Torino. Seeking a Master's Thesis or internship opportunity in biomedical applications, embedded systems and innovative healthcare technologies, with a strong interest in multidisciplinary engineering projects and product development.

## EDUCATION

**Double Master's Degree** Politecnico di Torino  
2025 – Present Turin, Italy  
• Completing final M.Sc. year in Biomedical Engineering.  
• **Current Average:** 26.6/30

**Biomedical Engineering Degree** National University of Córdoba  
2020 – 2024 Argentina  
• University Bachelor's Degree in Biomedical Engineering.

## TECHNICAL SKILLS

Biomedical Signal Processing

Embedded SystemsMicrocontrollers

Analog ElectronicsAmplifiersFilters

PythonMATLABCC++

## PROFESSIONAL SKILLS

Analytical ThinkingResearch Mindset

Independent WorkingProblem Solving

LeadershipTeam Collaboration

## LANGUAGES

Spanish: Native  
English: Level B2 (Certified)  
Italian: Level B2 (Certified)  
German: Level B1 (Certified / Currently improving)  
Mandarin Chinese: Basic

## EXPERIENCE

**Supervised Professional Internship** Conci Carpinella Institute  
May 2024 – Oct 2024 Córdoba, Argentina  
• Acquired and preprocessed neurophysiological signals using a 128-channel EEG system, including subject preparation, electrode placement, impedance verification, signal quality assessment, and acquisition software.  
• Participated in functional MRI (fMRI) studies, supporting experimental protocols and synchronization of task paradigms with MRI acquisition.

**President (2025–2026) & VP (2024)** IEEE UNC Student Branch  
Feb 2024 – Feb 2026 Córdoba, Argentina  
• Led multidisciplinary engineering teams, coordinating technical workshops, research initiatives, and collaborations with academia and industry.

**Multidisciplinary Innovation Program** iTeams – CONICET – Ministry of Science  
Aug 2024 – Oct 2024 Córdoba, Argentina  
• Evaluated technology transfer strategies to translate biomedical research into real-world healthcare applications.

**Course “AI Tools in Medical Imaging”** EMBS – IEEE / SABI  
June 2024 Argentina  
• Applied machine learning techniques for medical imaging analysis and computer-aided diagnosis.

**Collaboration in In Vivo Experiments** INICSA Institute  
May 2024 – Oct 2024 Argentina  
• Contributed to in vivo experimental protocols, biomedical data collection, and research on diabetes-related complications.

**International Fellowship & Clinical Internship** Friedrich Schiller Gymnasium & Pediatric Clinic  
Sep 2016 – Jul 2017 Marbach am Neckar / Bietigheim-Bissingen, Germany  
• Completed an international academic program, strengthening intercultural communication skills and German language proficiency.  
• Gained first-hand exposure to the German healthcare system through a clinical internship in a pediatric hospital.

## RELEVANT ENGINEERING PROJECTS

- Neural Signal Processing Using MEA Data:** Developed and validated a Python pipeline to process extracellular neuronal recordings from MEA datasets, implementing Butterworth filtering, spike/burst detection, feature extraction, and quantitative analysis to characterize network activity under developmental and pharmacological conditions.
- ECG Monitoring and Analysis Platform:** ECG Monitoring and Analysis Platform: Developed a Python telemedicine platform for visualization and analysis of multi-lead ECG recordings, applying digital signal processing techniques for R-peak detection, RR interval analysis, and heart-rate estimation from physiological signals.
- Embedded Brushless Motor Control System:** Developed an embedded control system in C using a PIC16F887 microcontroller, integrating a 4x4 matrix keypad, LCD interface, PWM motor speed control, and interrupt-driven RPM monitoring.
- Digital Twin for Remote Patient Monitoring:** Designed the architecture and user interfaces of a digital health platform integrating physiological sensor data, patient-reported outcomes, and clinical dashboards to support remote patient monitoring.



PORTFOLIO: milagroslahuerta.github.io  
Additional projects, certificates and supporting material available online.